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Down, set, record: Assessing warm-up structure and use of neuromuscular training exercises in Canadian adolescent tackle football

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Abstract

Implementing injury prevention strategies is critical for coaches and sport organizations to promote player safety. Neuromuscular training (NMT) warm-up programs that involve aerobic, strength, agility, balance/coordination, and head-and-neck control exercises have demonstrated injury rate reductions of >35% across many adolescent sports. Tackle football (i.e., American football) has amongst the highest injury rates within youth sport and lacks examination of current warm-up structure and exercises. Our video-analysis study examined footage from 23 practice and game warm-ups (46 total) to document the duration and exercises completed for nine adolescent (males, ages 14–17) tackle football teams over a 9-week season within Calgary (Canada). We found that teams typically spent less than five minutes actively warming-up with most exercises being aerobic (51%) and dynamic stretching (29%), while <10% of exercises involved static stretching (6%) or NMT components of strength (9%), agility (3%), balance/coordination (1%), and head-and-neck control (1%). Warm-up length and use of NMT components were similar between games and practices, and over the season. Our findings demonstrate that NMT exercises are not being used in tackle football, which supports future implementation and evaluation of a tackle football-specific NMT program. We also provide coaches with a method for examining their team's warm-up structure.

Keywords

Agility, Concussion, head-and-neck control, injury prevention, strength, stretching

Introduction

Tackle football (also known as American football) is a high intensity collision-based sport with over 1.6 million youth registrants under 18 years old each year in North America (Canada and United States).[1,2] Within the Canadian province of Alberta, an estimated 16% of high school boys (ages 14–18) report participating in tackle football each year.[3] Participation in tackle football involves players making explosive running, cutting, and change-of-direction movements while engaging in frequent player-to-player collisions. The high intensity and highly

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physical gameplay require players to have developed attributes of muscle strength, agility, balance/coordination, and head-and-neck control to not only be an effective player but to fend off musculoskeletal and concussion injuries.[4,5] Injury rates in tackle football are amongst the highest for all youth team collision-based sports with 1 in 4 adolescent players expected to suffer an injury each year contributing to an overall injury rate of 4.0–11.7 injuries per 1000 athletic exposures (game and practice sessions; AE) with 69% of injuries being muscle or joint/ligament sprains/strains (43% of injuries) and concussions (26% of injuries).[6–8] Due to the high risk of injury in tackle football, it is important that coaches and sport organizations seek to implement effective injury prevention strategies that not only prevent injuries but also target the development of a player's fitness attributes (i.e., strength, aerobic, balance, agility, head-and-neck control).

Neuromuscular training (NMT) programs have been developed to prevent injuries through targeting muscle activation, increased blood flow, joint stability, and mobility to prepare an athlete for the demands of sport.[9–11] For many adolescent sports, coaches are tasked with developing a conditioning plan for the season which typically includes deciding how long their players should warm-up and which exercises they should complete prior to games and practices. The objectives of a pre-session warm-up are to incorporate exercises that enhance an athlete's performance while participating in a sport or activity.[12] Fortunately, most NMT programs have been developed specifically for pre-session warm-up periods (e.g., 11+/FIFA 11+,[13] Rugby Activate,[5] SHRed Concussions[4]) and current evidence demonstrates that a 10 to 15 min NMT warm-up program that includes exercises from the fitness components of aerobic, strength, agility, and balance/coordination is effective for reducing the rate of musculoskeletal injuries by more than 35%.[4] More recently, a rugby-specific NMT program within adolescent rugby union players that included an additional component of head-and-neck control to promote proper posture and head positioning while tackling showed significant 59% lower rates of concussion when the program was completed ≥3 times per week.[5] Although it is unclear for the exact mechanism of concussion prevention through NMT, it may be a result of the program as a whole preparing the athlete's awareness for the demands of the sport or that the head-and-neck control component was similar to vestibular cervical rehabilitation exercises, which are postulated to protect against concussion and whiplash injuries.[14] While the current recommendations are to complete NMT warm-up programs 2–3 times per week to incur benefits, the dose-response relationship to completing the program along with the contribution of completing each individual component for injury prevention is not fully understood.[5,11,14]

Although there have been multiple sport-specific NMT warm-up programs developed and evaluated (e.g., 11+ Soccer Program, Rugby Activate Program, SHRed Injuries Basketball Program), there has been no evaluation of a tackle football-specific program.[5,15,16] Adherence is one of the main and most significant limitations to the effectiveness of a NMT warm-up program with only 27% of youth NMT evaluation studies having a moderate-to-high adherence (based on completing ≥1.5 times per week or cumulative program use of ≥70%); therefore, it is indeed important to develop programs that are useful and appealing to sport participants for completion.[17] Slauterbeck and colleagues (2017) examined the adherence and use of the 11+ (formerly FIFA 11+) NMT warm-up program – a program that emphasizes lower-extremity exercises – across multiple high school sports including tackle football in the United States.[18] They reported that program adherence was low with teams completing the program a mean of 1.45 times per week with exercises consisting mainly of running, dynamic mobility, and stretching.[18] As tackle football involves both the lower extremity (e.g., change-of-direction, evasive movements) and upper extremity (e.g., tackling, catching/throwing), it would be important to examine current warm-up structure and exercises that tackle football teams are completing to help inform future implementation of a tackle football-specific program. Therefore, the objective of this video analysis study was to examine practice and game warm-up footage for adolescent (ages 14–17, males) tackle football teams within Calgary (Canada) to document the warm-up length, the exercises completed, the utilization of the different NMT components, and to determine if warm-up structure differed between game and practice sessions and at different time points of the nine-week season.

Methods

Study design and participants

This cross-sectional video-analysis study is a sub-study within the SHRed Concussions (Surveillance in High Schools and Community Sport to Reduce Concussions and Their Consequences) research program. We filmed adolescent (ages 14–17) tackle football practice and game team warm-ups throughout the 2022 competitive season. Prior to filming, we obtained league and coach approval from all nine tackle football teams within the Calgary Spring Football Association (Calgary, Alberta, Canada) and this study received ethics approval from the University of Calgary's Conjoint Health Research Ethics Board (CHREB ID: REB21-0968).

Procedure

Structured practice and game warm-up sessions were filmed by trained research assistants throughout the season using a single camcorder (Panasonic *HC-V180K*) set-up on a tripod at field-level while ensuring all players were in the camera's

Table 1. Warm-up component descriptions.

Component	Description
Aerobic	A component of neuromuscular training that involves exercises aimed to increase heart rate and respiratory rate for increased blood flow to the extremities. This typically takes between 2–5 min to induce perspiration and a global cardiovascular response. **e.g.,** *Forward/Backward Running, Power Skips*
Strength	A component of neuromuscular training that involves exercises that typically involve eccentric loading in both open and closed kinetics chains, but can include isometric holds. The aim is to improve loading capabilities and tolerance of primary muscle groups (agonists, antagonists, and synergist). **e.g.,** *Nordic Hamstring Curls, Front Plank*
Agility	A component of neuromuscular training that involves exercises that include plyometrics, jumping, landing and change of direction though open chain kinematics to train the muscles, tendons, and nervous system. These exercises should be used and progressed with precision and speed to replicate sport competition demands and reaction skills. Emphasizing the importance of landing mechanics and technique is important as acute ankle and knee injuries can occur during landings. **e.g.,** *Plant and Cut or Change of Direction, Multidirectional Jumps*
Balance/ Coordination	A component of neuromuscular training that involves exercises that emphasize motor control, sensorimotor feedback, and proprioception for reacting in closed chain situations. Suggested to improve posture and dynamic balance. **e.g.,** *Airplane Hinge, Single Leg Hops*
Head & Neck Control	A component of neuromuscular training that involves exercises focused on improving cervical spine muscular control, endurance, strength, and head-on-neck position. These exercises may have significant sport benefits which include proper contact, anticipation contact and posture during play. As the newest component to neuromuscular training, clinical tests have suggested that decreases in cervical spine function may be a risk factor for concussion and aspects of rehabilitation. **e.g.,** *Cervical Spine Flexion/Extension, Static Neck Contractions*
Dynamic Stretching	Non-neuromuscular training exercises that involve light movement of targeted muscle groups in a controlled motion to improve muscle flexibility, range of motion, and joint mobility in preparation for activity. **e.g.,** *Leg Swings (hamstring), Arm Circles (shoulder)*
Static Stretching	Non-neuromuscular training exercises that involve being still and holding a muscle in the furthest comfortable lengthened position for a select period of time for promoting muscle flexibility and range of motion. **e.g.,** *Toe Reaches (hamstring), Overhead Arm Pull (triceps)*

Notes: Neuromuscular training components are shaded in light grey.

field of view. Video-analysis of warm-up footage was performed using Dartfish video analysis software (version 10.1). Dartfish allowed us to create a custom tagging panel to code each warm-up video based on length (seconds), NMT exercise components (i.e., aerobic, strength, agility, balance/coordination, head-and-neck control exercises) or non-NMT exercise components (i.e., dynamic stretching, static stretching), and to document specific exercises (e.g., lunges, sprints, high knees) for both practice and game warm-ups. Exercise components were defined based on the descriptions presented in Table 1. The tagging panel was created by both a Certified Strength & Conditioning Specialist (JTC) with experience prescribing exercises to collegiate and youth athletes, and a Certified Exercise Physiologist (CAV) who had over 7 years of experience as a NMT knowledge broker to community sports teams. The primary author (JTC) analyzed all warm-up videos after achieving ≥90% agreement with the NMT expert (CAV; reference standard) for identifying specific exercises and their corresponding component for 5 randomly selected warm-up session.

Statistical analysis

Stata version 17.0 (StataCorp, 2023) was used for all statistical analyses. Descriptive statistics reported the median time spent during warm-up (with 25th and 75th percentiles) and frequencies (with percentages) of exercises and components used. A full factorial mixed effects linear regression model was used to calculate means and mean differences (adjusted for cluster by team as a random effect) for time spent within each exercise component (e.g., aerobic exercises, dynamic stretching) between session type (game or practice) and season period (beginning/weeks 1–3, middle/weeks 4–6, and end/weeks 7–9). The model used restricted maximum likelihood estimation and Kenward-Rodger degrees of freedom to adjust for the small number of teams (clusters) in the analysis to

produce average marginal effects presented in the results.[19] Statistical significance ($p < 0.05$) of average marginal differences was determined through the Delta method (denoted by Δ) and aggregated total times were computed through the summation of linear combinations to produce average marginal means from the full factorial model. In effect, a single multivariable model was used to produce all the comparisons of warm-up structure by session type and season period.

Results

Overall, there were 46 warm-ups filmed and analyzed (n = 23 game, n = 23 practice) from nine community tackle football teams during the nine-week 2022 season (April to June 2022). Each team had a minimum of four sessions (two practices and two games) filmed over the season, and a median of five warm-ups were filmed per team (range: 4–6 warm-ups). The median time that teams allocated to warming-up was just over seven-and-a-half minutes (median = 456.9 s, 25th–75th percentile; 276.8–532.3 s) and players spent a median time of nearly four-and-a-half minutes actively completing exercises (median = 266.7 s, 25th–75th percentile; 193.5–326.9 s). Table 2 displays the observed proportions of exercise components used by the teams and the observed median time (seconds) spent within each warm-up component stratified by session type and season time-period.

Representation of the different exercise components varied over the 46 warm-ups, with at least one aerobic exercise being included in 98% of warm-ups (45/46), dynamic stretching being included in 91% of warm-ups (42/46), and strength exercises being included in 76% of warm-ups (35/46). All other components were represented in less than 20% of warm-ups, and included agility (7/46 warm-ups; 15%), head-and-neck control (6/46 warm-ups; 13%), static stretching (5/46 warm-ups; 11%), and balance/coordination (3/46 warm-ups; 7%). Moreover, the most utilized exercises across all warm-up sessions were aerobic exercises (283/550 exercises; 51%) and dynamic stretching (158/550 exercises; 29%) components, while the strength, agility, head-and-neck control, and static stretching components were used for <10% of all exercises (Table 3). Dynamic and static stretching exercises are not considered NMT components but collectively made up 192/550 (35%) of exercises completed (Table 3). The most common exercises completed in each component were skips (aerobic exercise, 76/283 exercises; 27%), variations to lunges (strength, 50/51 exercises; 98%), deceleration and change of direction (agility, 10/15 exercises; 66%), single leg hip hinge (balance/coordination, 2/3 exercises; 67%), and bear crawl bobbleheads (head-and-neck control, 6/6 exercises; 100%) (Table 3). The hips were most targeted for both dynamic (90/158 exercises; 57%) and static stretching (25/34 exercises; 74%) (Table 3).

Results from the full factorial mixed effects linear regression model demonstrated no differences between practices and game sessions for the mean time spent warming-up overall and for each specific NMT and stretching component (while adjusting for season period) (Table 4). For both game and practice warm-ups, most time was spent completing aerobic exercises (mean = 118.5 s per warm-up, 95% CI; 102.7–134.3) and dynamic stretching exercises (mean = 93.8 s per warm-up, 95% CI; 78.0–109.6), while the least amount of time was spent completing agility exercises, balance exercises, and head-and-neck control exercises (Table 4).

When examining warm-up structure over different periods of the season (while adjusting for session type), the overall time spent warming-up did not differ for any time periods within the season (Table 5). When examined further by specific warm-up components, the only differences found between season periods was for the aerobic component, where period 2 (weeks 4–6) had the longest mean time spent completing aerobic exercises, which was 42.2 s longer than period 1 (weeks 1–3; mean $\Delta = 42.2$ s, 95% CI; 5.8–78.7) and 66.4 s longer than period 3 (weeks 7–9; mean $\Delta = 66.4$ s, 95% CI; 28.4–104.5) (Table 5). There were no significant differences between time spent completing aerobic exercises for periods 3 and 1 and there were no other statistically significant differences for the time spent in any other components across all time-period comparisons (Table 5). Figures 1 and 2 descriptively display the stratum-specific mean times spent in warm-up components by time-period for both practice (Figure 1) and game sessions (Figure 2). Due to limited sample size in many of the stratum-specific components, the figures are meant to be descriptive which precluded any analytical comparisons.

Discussion

This study used video-analysis to describe the warm-up structure, exercises completed, and use of NMT and non-NMT components for warm-ups over the course of an adolescent Canadian tackle football season. We found that teams allocated a median time of seven-and-a-half minutes (456.9 s) to warming-up and players spent a median time of nearly four-and-a-half minutes (266.7 s) actively completing exercises. As the current recommendation for NMT program effectiveness is to complete a 10 to 15-min warm-up, our findings suggest that coaches are not allocating enough time for players to warm-up. Moreover, with a considerable amount of time (median of 3 min) being spent inactive during warm-up to transition between exercises, we recommend that tackle football teams revisit their exercises and warm-up flow to maximize active warm-up time.

We found that aerobic exercises and dynamic stretching accounted for 80% of all exercises completed, and that the

Table 2. Description of observed warm-up components across the season time periods and stratified by session type.

	Season Period 1 (Weeks 1–3)				Season Period 2 (Weeks 4–6)				Season Period 3 (Weeks 7–9)			
	Game		Practice		Game		Practice		Game		Practice	
Component	Median Time (seconds) [IQR]	% of teams (n)	Median Time (seconds) [IQR]	% of teams (n)	Median Time (seconds) [IQR]	% of teams (n)	Median Time (seconds) [IQR]	% of teams (n)	Median Time (seconds) [IQR]	% of teams (n)	Median Time (seconds) [IQR]	% of teams (n)
Overall	229.0 (178.6–238.5)	55% (5/9)	279.0 (258.6–383.6)	55% (5/9)	272.3 (215.8–290.6)	66% (6/9)	316.0 (229.8 416.8)	100% (9/9)	325.5 (195.4–362.7)	100% (9/9)	279.0 (258.6–383.6)	33% (3/9)
Aerobic	87.7 (73.3–136.4)	60% (3/5)	97.9 (80.4–153.7)	80% (4/5)	116.1 (82.0–145.9)	66% (4/6)	120.7 (64.1–178.2)	100% (9/9)	86.9 (59.0–157.1)	100% (9/9)	69.4 (26.0–85.2)	100% (9/9)
Strength	43.2 (26.4–64.9)	80% (4/5)	33.8 (30.7–55.7)	80% (4/5)	44.4 (38.6–45.4)	66.6% (4/6)	41.4 (29.1–67.5)	77% (7/9)	33.9 (30.1–67.1)	77% (7/9)	49.3 (17.7–86.0)	100% (9/9)
Agility	[a]67.3 (67.3–67.3)	20% (1/5)	[a]138.8 (138.8–138.8)	20% (1/5)	0	0% (0/6)	19.0 (15.9–32.5)	11% (1/9)	[a]26.8 (26.8–26.8)	11% (1/9)	0	0% (0/9)
Balance	0	20% (1/5)	[a]9.0 (9.0–9.0)	0% (0/5)	[a]26.7 (26.7–26.7)	16.6% (1/6)	16.7 (15.1–18.3)	22% (2/9)	0	0% (0/9)	0	0% (0/9))
Head & Neck Control	[a]46.0 (46.0–46.0)	20% (1/5)	0	0% (0/5)	[a]40.3 (40.3–40.3)	16.6% (1/6)	56.7 (48.2–65.1)	11% (1/9)	50.0 (34.4–65.7)	11% (1/9)	0	0% (9/9)
Dynamic Stretching	78.9 (60.2–93.1)	80% (4/5)	164.4 (114.5–199.2)	100% (5/5)	84.4 (72.3–97.8)	100% (6/6)	109.5 (89.5–125.3)	100% (9/9)	95.8 (79.3–110.7)	88% (8/9)	82.0 (49.5–106.9)	100% (9/9)
Static Stretching	0	0 (0/5)	0	0% (0/5)	[a]116.6 (116.6–116.6)	16.6% (1/6)	188.0 (140.6–190.9)	11% (1/9)	[a]109.0 (109.0–109.0)	11% (1/9)	0	0% (0/9)

Notes: Neuromuscular training components are shaded in light grey. Median time in each component time is restricted to teams who completed at least one exercise in that component. Proportions of component use are based on the number of teams that were videotaped during each season period which are shown in the overall row (for example, all 9 teams were filmed for at least one practice warm-up in season period 2 and game warm-up for season period 3).
[a]Single warm-up observation for a component.

Table 3. Description of the exercises observed in each of the components.

Component and exercise	No. of exercises (%)	Exercise resembles NMT?
Overall	**550 (100%)**	**n/a**
Aerobic	**283 (51%)**	**Yes**
Skips	76 (27%)	Yes
Forward/Backward Run	48 (17%)	Yes
Heel Kicks	43 (15%)	Yes
High Knees	26 (9%)	Yes
Jumping Jacks	25 (9%)	Yes
Sprints	23 (8%)	Yes
Carioca	20 (7%)	Yes
Leg swings	11 (4%)	Yes
Side Shuffle (no change of direction)	11 (4%)	Yes
Strength	**51 (9%)**	**Yes**
Walking Lunges	35 (69%)	Yes
Lateral Walking Lunges	15 (29%)	Yes
Dynamic Squat	1 (2%)	Yes
Agility	**15 (3%)**	**Yes**
Deceleration	5 (33%)	Yes
Direction Change	5 (33%)	Yes
Quick Start/Stop	3 (20%)	Yes
Lateral Side Shuffle (change of direction)	2 (14%)	Yes
Balance/Coordination	**3 (1%)**	**Yes**
Hip Hinge Single Leg	2 (67%)	Yes
Heel Walks	1 (33%)	Yes
Head & Neck Control	**6 (1%)**	**Yes**
Bear Crawl & Bobbleheads	6 (100%)	Yes
Dynamic Stretching	**158 (29%)**	**No**
Hips	90 (57%)	n/a
Knee	15 (10%)	n/a
Shoulder	7 (4%)	n/a
Multiple	46 (29%)	n/a
Static Stretching	**34 (6%)**	**No**
Hips	25 (74%)	n/a
Knee	8 (23%)	n/a
Torso	1 (3%)	n/a

Notes: Neuromuscular training components are shaded in light grey. For better generalizability, dynamic and static stretching are coded as the targeted anatomical location for the stretches.

predicted mean time estimated from the multivariable model was approximately two minutes for aerobic exercises (118.5 s) and a minute-and-a-half for dynamic stretching

Table 4. Comparison of estimated time (seconds) spent in each component based on session type.

	Mean time (seconds) (95% CI)			Mean difference (seconds) (95% CI)	
Component	Overall	Practice	Game	Game – Practice	*p*
Overall	274.9 (224.0, 325.7)	276.5 (211.1, 341.9)	273.4 (208.4, 338.4)	−3.1 (−84.8, 78.7)	0.94
Aerobic	118.5 (102.7, 134.3)	132.0 (109.9, 154.1)	106.1 (84.3, 128.0)	−25.9 (−56.3, 4.6)	0.10
Strength	35.4 (19.6, 51.3)	32.5 (10.4, 54.6)	38.1 (16.3, 59.9)	5.6 (−24.8, 36.0)	0.72
Agility	6.4 (0[a], 22.2)	3.5 (0[a], 25.6)	9.01 (0[a], 30.8)	5.5 (−24.9, 36.0)	0.72
Balance	1.5 (0[a], 17.4)	1.7 (0[a], 23.8)	1.4 (0[a], 23.2)	−0.3 (−30.8, 30.1)	0.98
Head & Neck Control	5.4 (0[a], 21.2)	7.5 (0[a], 29.6)	3.5 (0[a], 25.3)	−4.0 (−34.5, 26.4)	0.80
Dynamic Stretching	93.8 (78.0, 109.6)	88.0 (66.0, 110.1)	99.1 (77.2, 120.9)	11.0 (−19.4, 41.5)	0.48
Static Stretching	13.8 (0[a], 29.6)	11.2 (0[a], 33.3)	16.2 (0[a], 38.0)	5.0 (−25.5, 35.4)	0.75

Notes: Neuromuscular training components are shaded in light grey. Mean times are marginal effects estimates from the full factorial mixed effects linear regression model, adjusted for season period. *Statistically significant finding ($p < 0.05$).
[a]All negative values in the mean time estimate and lower bound of the confidence interval were truncated to 0.

(93.8 s). Use of strength exercises were the third most common (9% of all exercises) with a mean of 35 s spent completing per warm-up; however, 98% of strength exercises completed were a variation of the lunge which would limit the activated muscle groups. For example, other NMT warm-up programs such as the 11+ (formerly FIFA 11+) include multiple strength exercises such as Nordic hamstring curls, side planks, and front planks to activate multiple muscle groups[16] which is important to recruit and activate a greater number of skeletal muscle groups for movement and stability.[20,21]

We found that coaches consistently implemented a structured warm-up throughout the season and that the time spent overall and in nearly all of the components did not differ between games or practices nor for the different time periods of the season. Our only statistically significant finding was that the aerobic component was 42 to 66 s longer in the season's middle compared with the beginning and end periods, respectively. However, with over half of the warm-up time and warm-up exercises being aerobic-related, we are uncertain if this short amount of time difference would be meaningful for injury prevention benefits or even increased aerobic capacity. Kilduff et al. (2023) reports that training adaptation occurs in response to both resistance and speed-based training rather than speed-based training alone (e.g., aerobic exercises), and when coupled with additional strength exercises, there are likely significant enhancements to performance.[22] In combination, these findings support that more variation in exercise components are required to maximize performance-based enhancements and injury prevention benefits.

Our findings of component utilization are similar to previous reports examining use of the 11+ (formerly FIFA 11 +) NMT program within United States high school sports, which found that tackle football teams included aerobic exercises within 91% of warm-ups (98% in our present study), strength exercises within 63% of warm-ups (76% in our present study), and agility/balance within 13% of warm-ups (15% agility, 7% balance/coordination in our present study).[18] Interestingly, we found a lower prevalence of static stretching (11% of warm-ups) compared with Slauterbeck and colleagues who reported 44% of warm-ups included static stretching during their examination of the 2015–2016 season, which may be a result of ongoing awareness and research findings that NMT and dynamic exercises provide more benefits to joint mobility and muscle performance than static stretching.[18,23,24] Agility, balance/coordination, and head-and-neck control components were the most underutilized NMT components found in this study, which demonstrates that tackle football coaches are missing an opportunity to include evidence-informed injury prevention exercises that help develop a range of player fitness attributes.[10,25,26] Moreover, as a newer component within the NMT literature, we found that only one team was using head-and-neck control exercises during their warm-ups consistently over the course of the season which is likely very applicable and important to help teach tackle football players proper body, neck, and head positioning for player-to-player contact. Indeed, proper tackling technique has been a considerable issue within tackle football due to concerns for fatalities, spinal injuries, moderate and severe brain injuries, as well as concussions.[27] As tackle football has the second highest overall concussion rate (behind rugby) and highest game concussions rate of all sports, it would indeed be crucial for coaches to introduce head-and-neck control exercises to promote neck strength building (e.g., via isometric contractions), as well as exercises that target proper heads-up posture positioning and cervical spine control.[5,28,29]

Table 5. Comparison of time (seconds) spent in NMT components based on period of the season (beginning, middle, end).

	Mean Time (seconds) (95% CI)			Mean Difference (seconds) (95% CI)					
Component	Period 1 (weeks 1–3)	Period 2 (weeks 4–6)	Period 3 (weeks 7–9)	Period 2 – 1	*p*	Period 3 – 1	*p*	Period 3 – 2	*p*
Overall	276.8 (191.1, 362.5)	309.5 (245.6, 373.3)	210.9 (108.3, 313.4)	32.7 (–66.8, 132.1)	0.52	–65.9 (–194.4, 62.6)	0. 32	–98.6 (–210.6, 13.5)	0.09
Aerobic	104.2 (74.1, 134.3)	146.5 (125.1, 167.9)	80.0 (47.9, 112.1)	42.2 (5.8, 78.7)	***0.02**	–24.2 (–67.9, 19.5)	0.28	–66.4 (–104.5, –28.4)	***<0.001**
Strength	36.3 (6.2, 66.4)	30.9 (9.5, 52.3)	42.8 (10.7, 74.9)	–5.4 (–41.9, 31.1)	0.77	6.5 (–37.2, 50.1)	0.77	11.9 (–26.1, 49.9)	0.54
Agility	21.7 (0[a], 51.8)	2.8 (0[a], 24.2)	0.8 (0[a], 32.9)	–18.9 (–55.4, 17.6)	0.31	21.0 (–64.6, 22.7)	0.35	–2.1 (–40.1, 36.0)	0.92
Balance	1.8 (0[a], 31.9)	2.6 (0[a], 24.0)	0[a] (0[a], 31.6)	0.8 (–35.6, 37.4)	0.96	–2.3 (–45.9, 41.4)	0.92	–3.1 (–41.1, 34.9)	0.87
Head & Neck Control	5.2 (0[a], 35.3)	6.1 (0[a], 27.5)	4.3 (0[a], 36.4)	0.9 (–35.6, 37.4)	0.96	–0.9 (–44.6, 42.7)	0.97	–1.8 (–39.9, 36.2)	0.92
Dynamic Stretching	106.7 (76.6, 136.8)	95.9 (74.5, 117.3)	80.1 (48.0, 112.2)	–10.7 (–47.2, 25.8)	0.56	–26.6 (–70.3, 17.1)	0.23	–15.9 (–53.9, 22.2)	0.41
Static Stretching	0.8 (0[a], 30.9)	24.6 (3.2, 46.0)	4.7 (0[a], 36.8)	23.8 (–12.7, 60.3)	0.20	3.9 (–39.8, 47.6)	0.86	–19.9 (–57.9, 18.2)	0.31

Notes: Neuromuscular training components are shaded in light grey. Mean times are marginal effects estimates from the full factorical mixed effects linear regression model, adjusted for session type. *Statistically significant finding ($p < 0.05$).

[a]All negative values in the mean time estimate and lower bound of the confidence interval were truncated to 0.

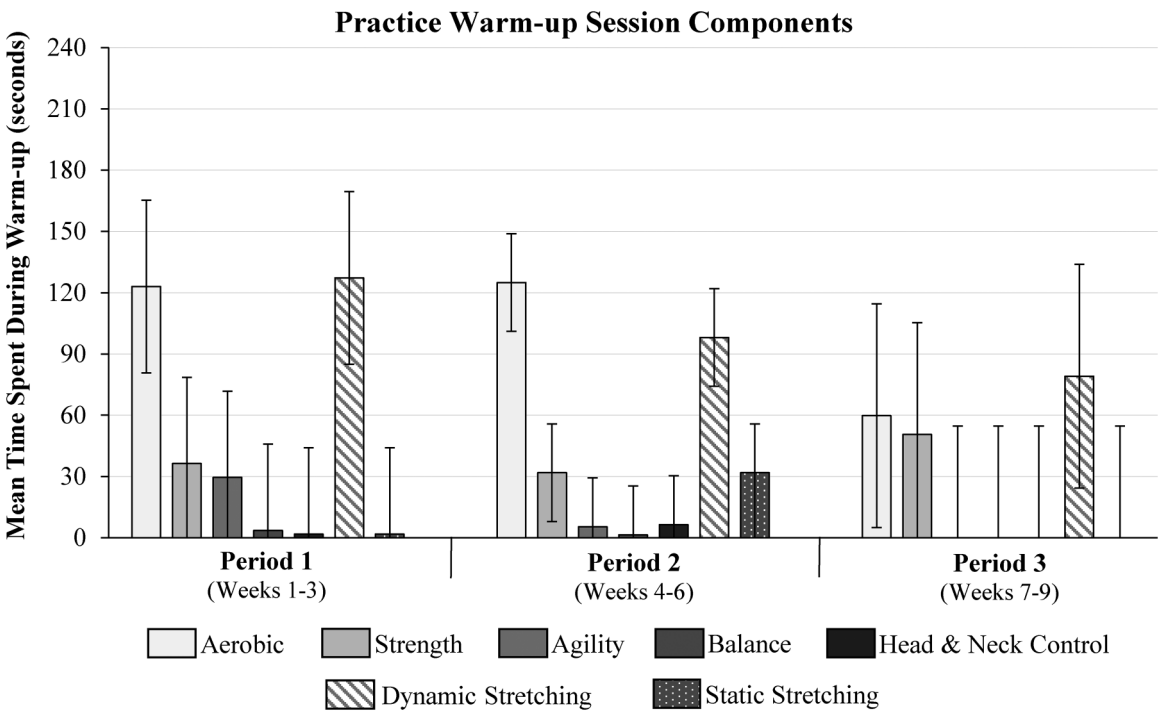

Figure 1. Time spent (seconds) completing warm-up components during practice sessions through different periods of the 9-week season.
Notes: Solid colours represent NMT Components. Patterned black and white shades represent dynamic and static stretching (non-NMT components). 95% confidence intervals are shown with mean time estimates (seconds). Negative confidence interval bounds were truncated to 0 when applicable.

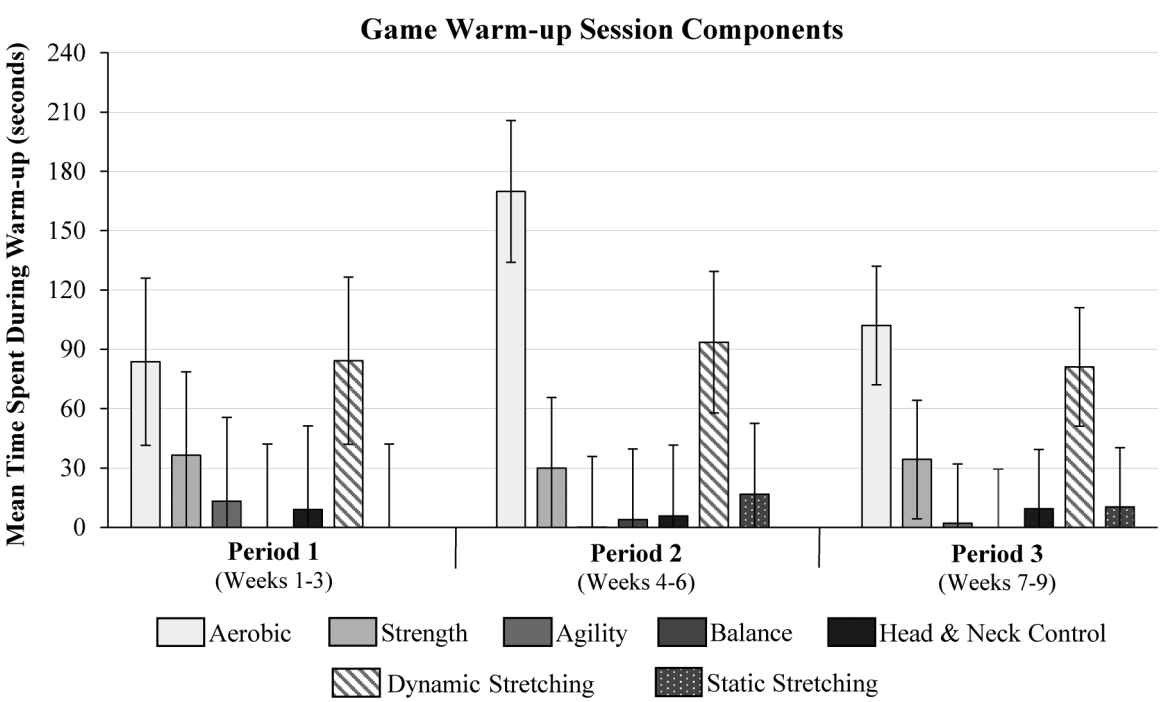

Figure 2. Time spent (seconds) completing warm-up components during game sessions through different periods of the 9-week season.
Notes: Solid colours represent NMT Components. Patterned black and white shades represent dynamic and static stretching (non-NMT components). 95% confidence intervals are shown with mean time (seconds). Negative confidence interval bounds were truncated to 0 when applicable.

The relatively inconsistent and low use of many NMT component exercises found in this study are likely due to previously documented barriers to NMT implementation and include coach and tackle football organization's lack of knowledge for NMT programs as a whole, having a perceived lack of time to implement a 10–15 min warm-up, skepticism for benefits from a NMT program, and lacking knowledge for how to properly complete exercises.[17,30,31] As adolescent tackle football teams can range from 35–80 players, it may also be difficult for coaches to monitor exercise fidelity amongst individual players which could hinder the use of new exercises. Our use of video-analysis would provide coaches with a method to examine their team's current warm-up structure, player exercise fidelity, and identify potential deficits and future opportunities to include injury prevention exercises. As no prior NMT program has been implemented within adolescent Canadian tackle football, our study was descriptive in nature to document typical warm-up structure and exercises that coaches are implementing with their teams as a means to educate coaches and tackle football organizations on strategies for future implementation and evaluation of a tackle football-specific NMT program.

Limitations

Limitations of this study include sample size (a minimum of 4 warm-ups filmed per team) which was based on feasibility for data collection for availability of both camcorders and research assistants. Having a limited number of warm-up recordings would ultimately lower the precision in our estimates; however, we do not believe this limitation would negate the structure of warm-ups, exercises and proportion of components completed, nor the time spent warming up. Second, as the research team received coach and league approval to film warm-ups throughout the season, this may have led teams to change their warm-up behaviours (e.g., allocating more time, completing more exercises) while in the presence of the research team to what they typically would complete. Although this change of behaviour was possible, as we did not introduce any intervention for warm-up programs nor provide any education modules for NMT warm-ups, we do not expect this to have been a significant limitation for this study's results. Moreover, we examined team structured warm-ups which would not include any self-directed or individual player warm-up exercises that were completed outside of the team warm-up. As it is unknown for the prevalence of players who complete individual warm-ups on their own, and typical exercises that they would complete, we are unsure how this limitation would affect the current study. Lastly, exercises were categorized into a single component (based on the primary author's determination while watching the warm-up exercise) and exercise fidelity was not examined. As some exercises may be considered to fit into multiple components depending on team instruction (e.g., side shuffles could be categorized as both aerobic and agility), the category that was most fitting was decided at the time of review by the primary author (JTC) based on his strength and conditioning expertise and how the team was completing the exercise. Not examining exercise fidelity limits our understanding for exercise form and the effort players are putting into their warm-up. As examining individual players during warm-up was not feasible for this study, future research that evaluates a NMT warm-up program should examine exercise fidelity while providing coaches with a practical session and resources for correctly teaching and monitoring exercise technique for their players.

Implications

In summary, our results suggest there is significant opportunity for tackle football coaches to improve their team warm-up structure in both ensuring exercises from the different NMT domains are represented and allocating more time for players. As half of the time allocated to warm-up was spent being inactive and transitioning between exercises, coaches should seek to improve the flow of their warm-up protocol to ensure players are spending 10–15 min actively engaging in exercises. Moreover, as proper heads-up tackle technique is an increasingly important avenue for concussion prevention, including head-and-neck control exercises within warm-ups would indeed be key for player safety. Lastly, as many of the NMT components of agility, balance/coordination, and strength were underutilized compared with dynamic stretching and aerobic exercises, educational resources would be important to provide coaches with diverse exercises to target tackle football-specific skills. The present study is important to provide coaches across sports with a method to examine their current warm-up protocol for identifying potential avenues to incorporate evidence-informed injury prevention exercises for their players.

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Consent to participate

We obtained league and coach approval from all nine tackle football teams within the Calgary Spring Football Association (Calgary, Alberta, Canada) with players who had consented to participate in the SHRed Concussions Research Program that this sub-study was part of.

Consent for publication

Not applicable.

Data availability

Available upon reasonable request.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical considerations

This study received ethics approval from the University of Calgary's Conjoint Health Research Ethics Board (CHREB ID: REB21-0968).

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